Structure of Molecules

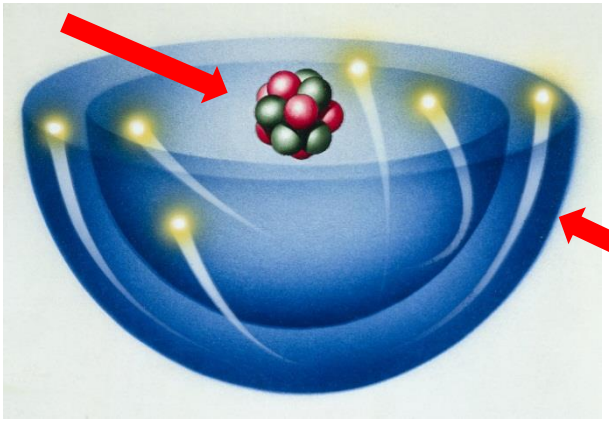
- How we know that electrons have different energies?
- How electrons fit into atomic orbitals?
- How atomic orbitals combine to make molecular orbitals?
- Why organic molecules have linear, planar, or tetrahedral structures?
- Connection between shape and electronic structure
- A true system of molecular orbital energies for simple molecules
- Why such rigor is not possible for typical organic molecules?

Why and How?

- **How does it feel to be a molecule?**
- **How atoms are held together?**
- **What is a chemical bond?**
- **Why molecules have particular shapes?**

Atomic Structure

- protons and neutrons

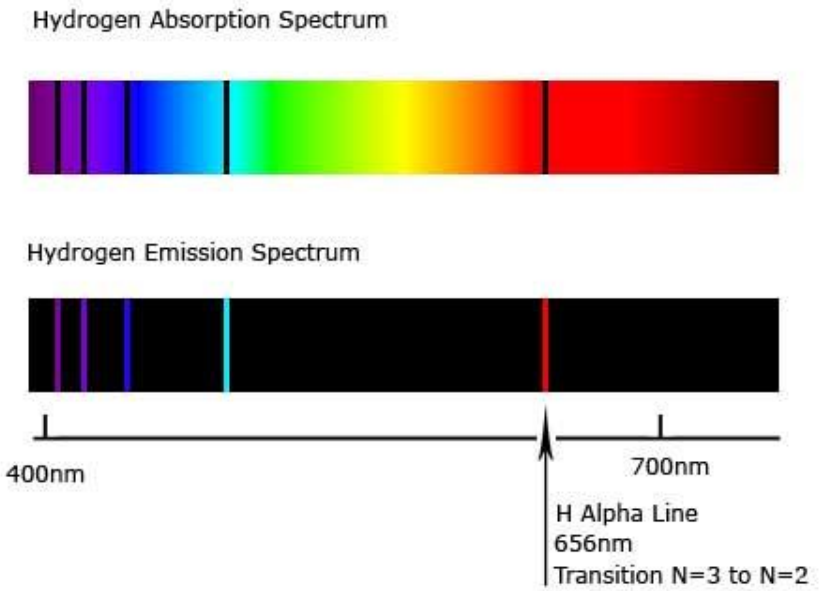


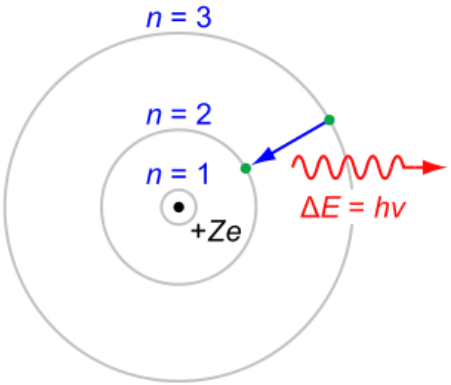
electron cloud

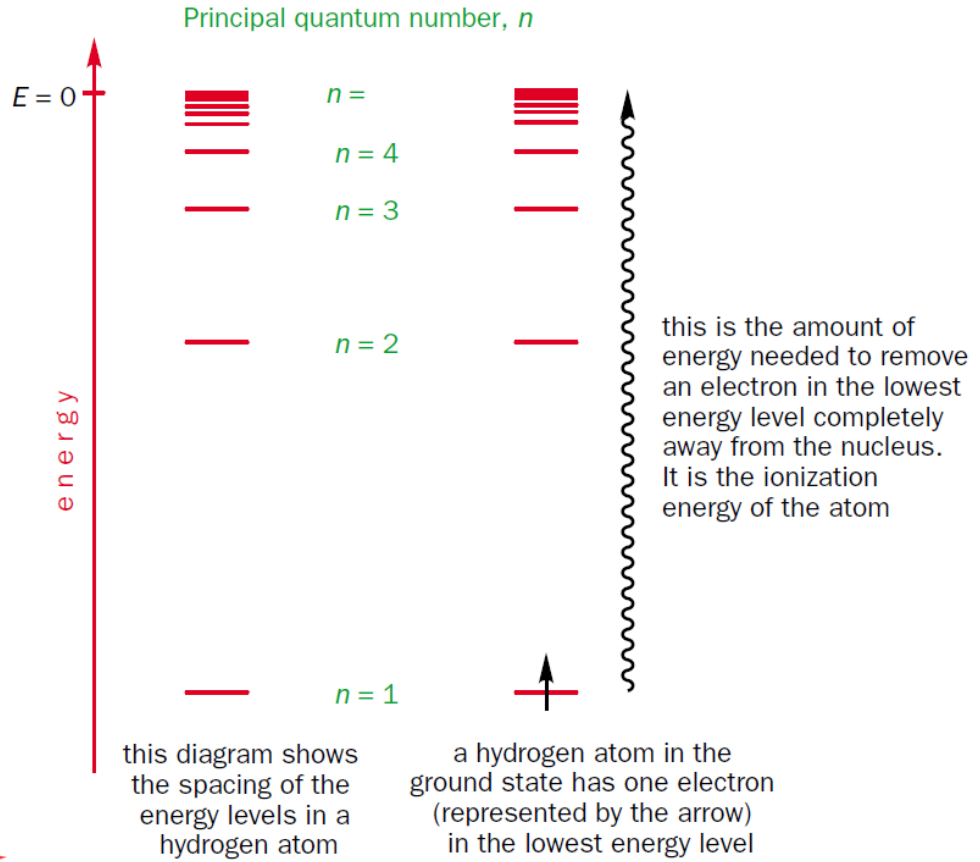
- The nucleus contains positively charged protons and uncharged neutrons
- The electron cloud is composed of negatively charged electrons

"Wave-particle duality": Electrons exhibit both particle and wave properties

How we know that electrons have different energies?





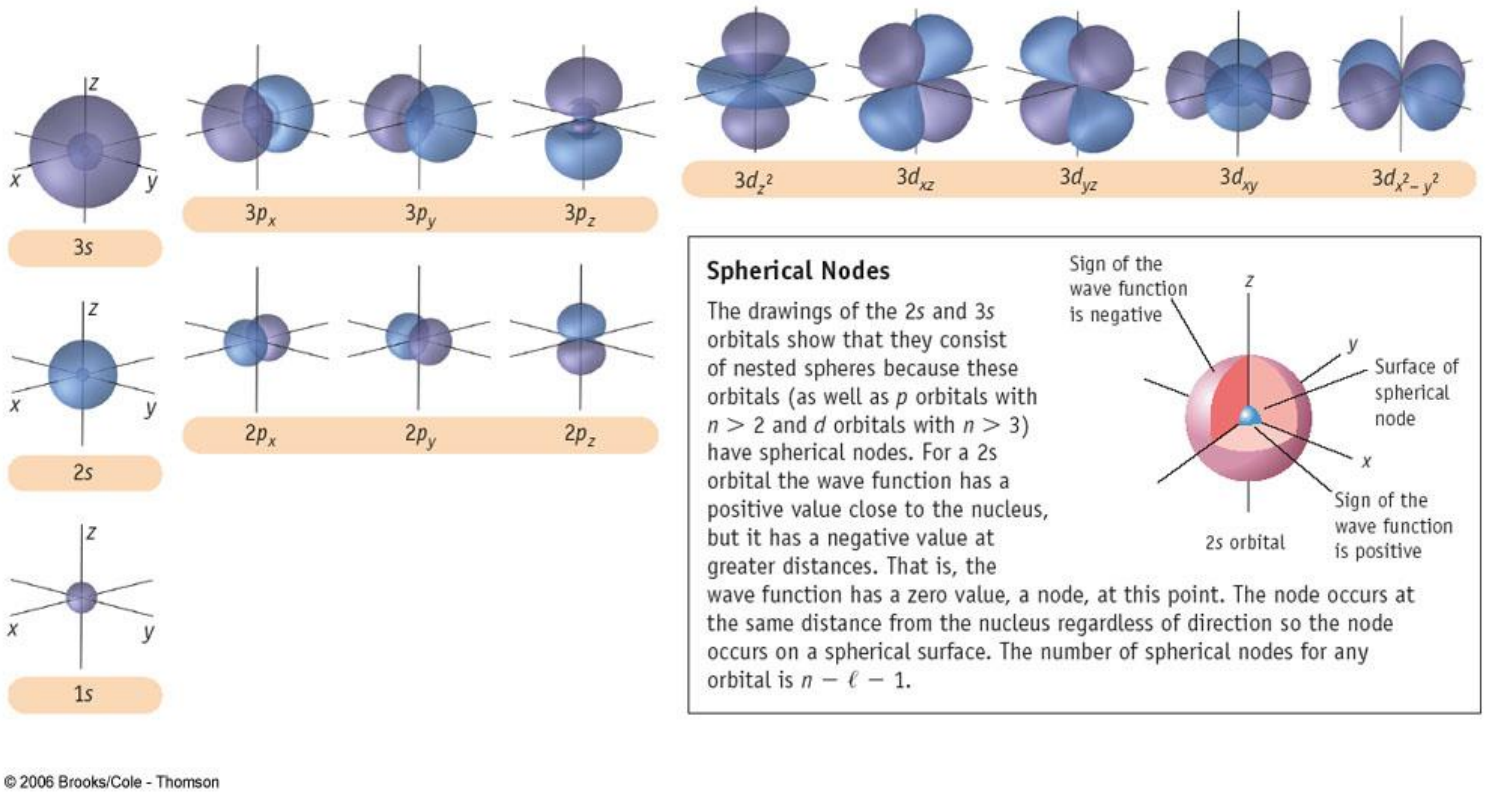


Electronic Energy Levels



- **Electrons can only exist in quantized energy levels-they cannot climb half a step!**
- **Electrons can be found in *atomic orbitals***

What are atomic orbitals?



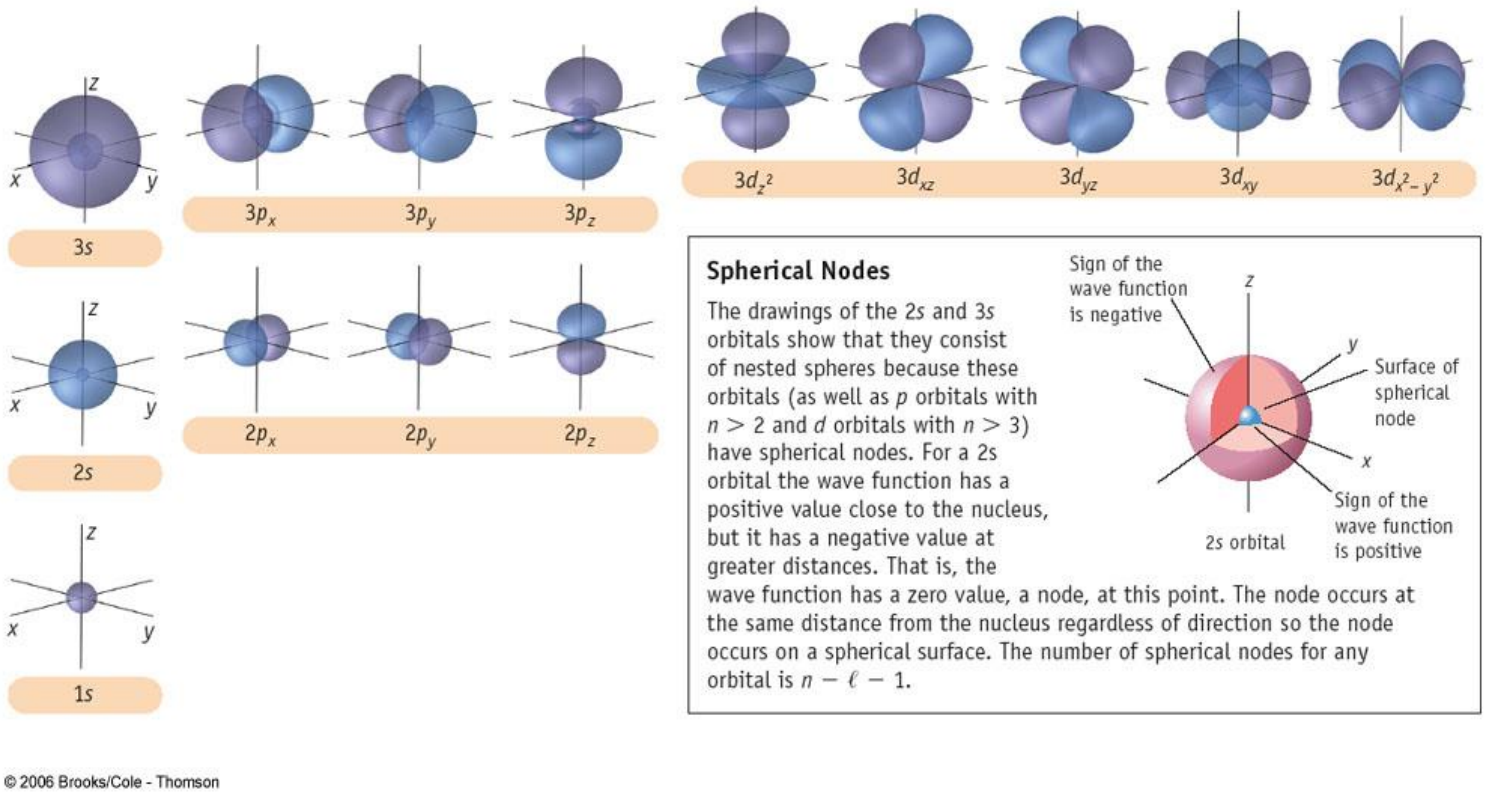
Spherical Nodes

The drawings of the 2*s* and 3*s* orbitals show that they consist of nested spheres because these orbitals (as well as *p* orbitals with $n > 2$ and *d* orbitals with $n > 3$) have spherical nodes. For a 2*s* orbital the wave function has a positive value close to the nucleus, but it has a negative value at greater distances. That is, the wave function has a zero value, a node, at this point. The node occurs at the same distance from the nucleus regardless of direction so the node occurs on a spherical surface. The number of spherical nodes for any orbital is $n - \ell - 1$.

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- **Atomic Orbitals (AOs) are wave functions that represent the region in space around the nucleus where electrons are having a probability to be found (Heisenberg's uncertainty principle)**

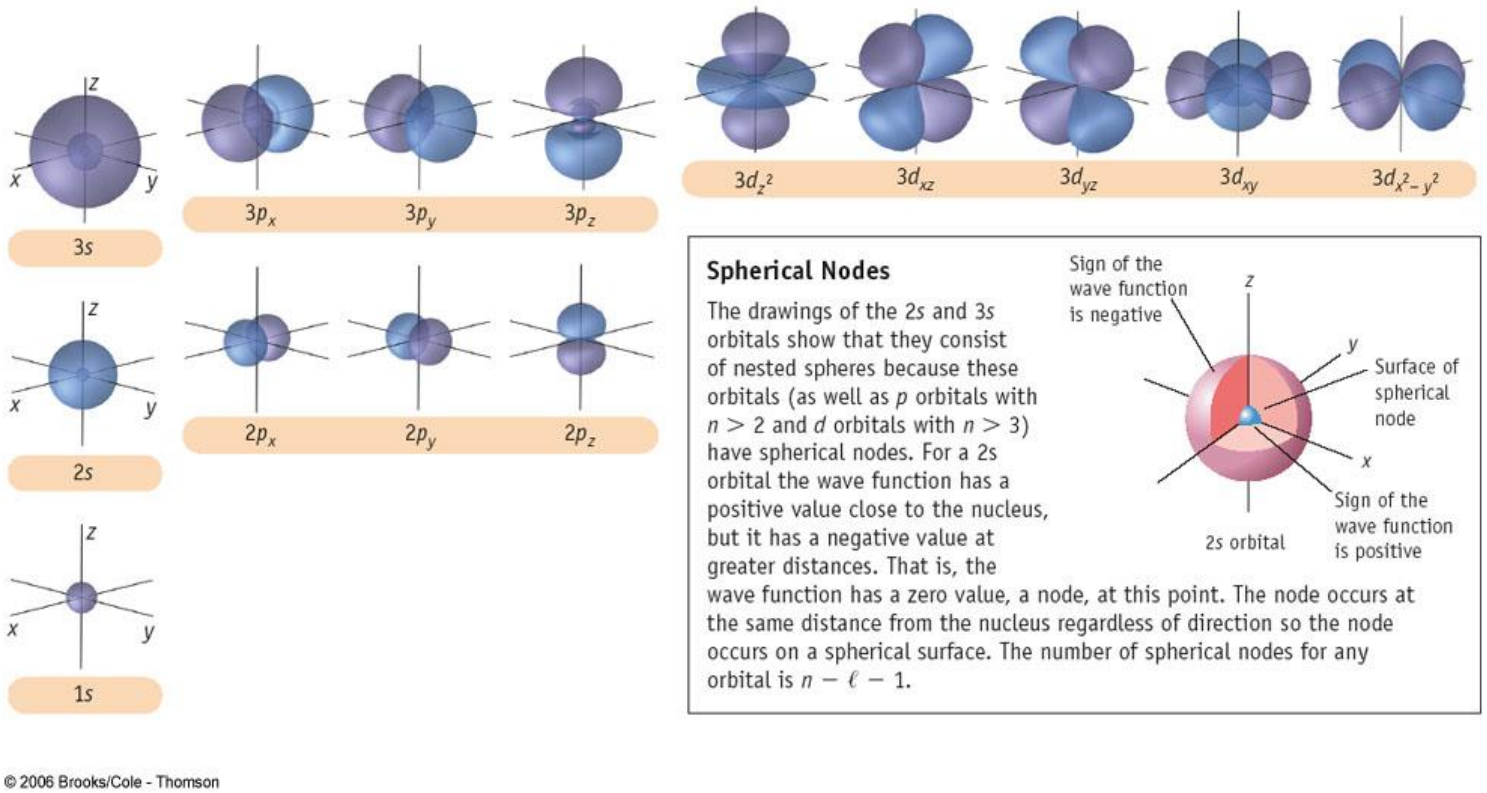
What are atomic orbitals?



© 2006 Brooks/Cole - Thomson

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- **Up to 2 electrons can be found in an AO, with opposite spin (Pauli exclusion principle)**

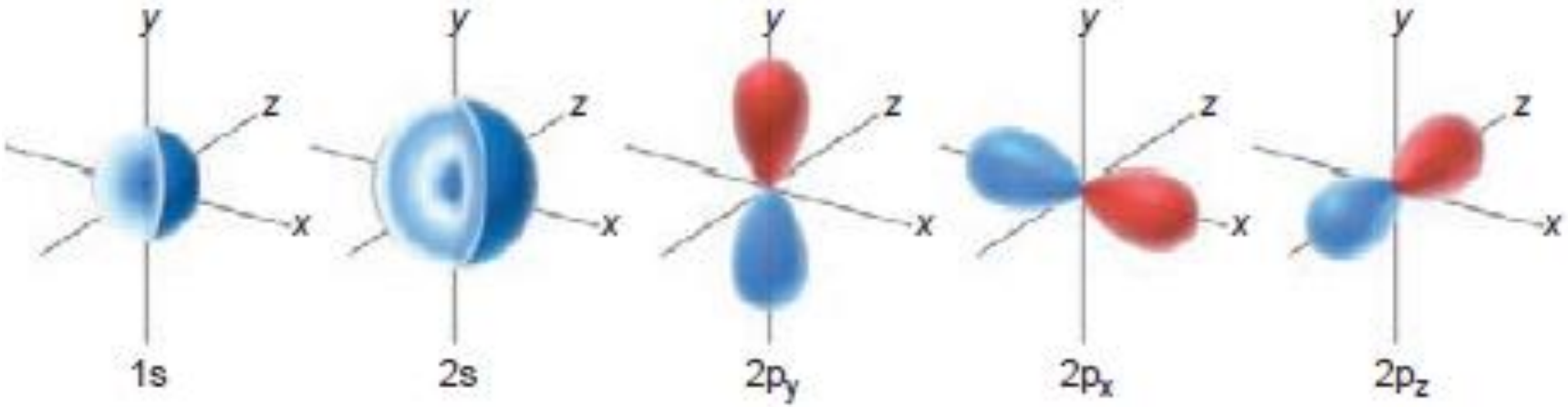
What are atomic orbitals?



© 2006 Brooks/Cole - Thomson

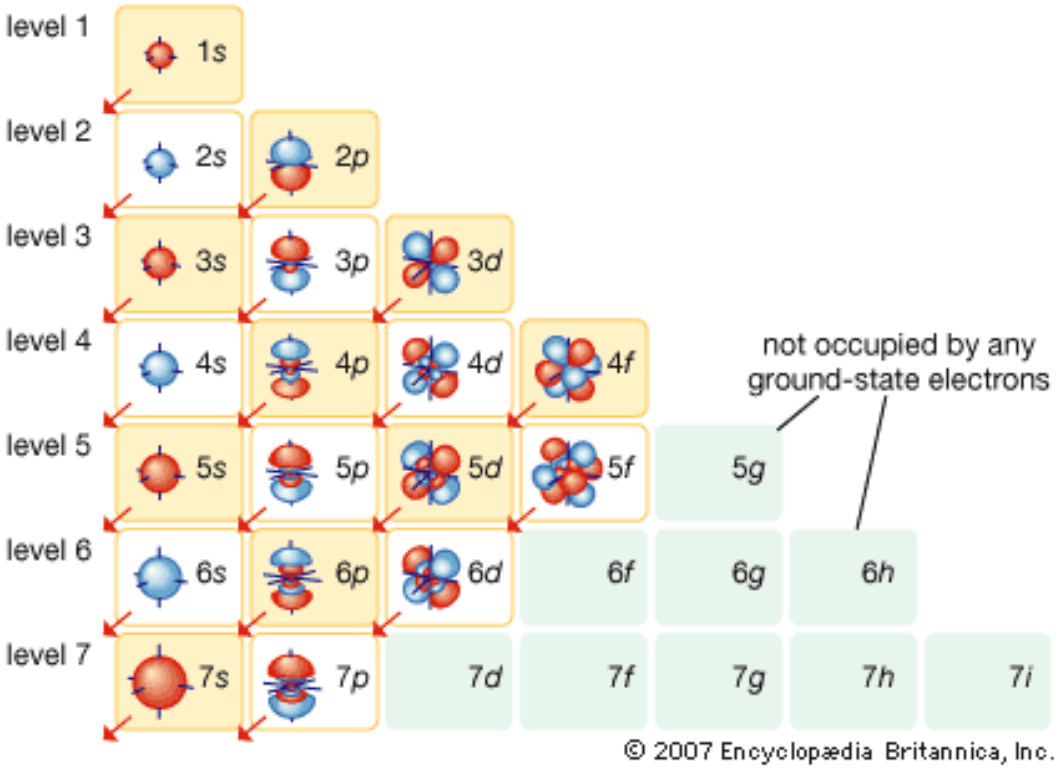
- **Atomic Orbitals (AOs) are wave functions that represent the region in space around the nucleus where electrons are having a probability to be found (Heisenberg's uncertainty principle)**
- **Up to 2 electrons can be found in an AO, with opposite spin (Pauli exclusion principle)**
- **Electrons prefer to be unpaired if they can occupy AOs at the same energy level, i.e. degenerate AOs (Hund's rule)**

1s, 2s and 2p atomic orbitals



The Aufbau principle

A hypothetical process in which an atom is "built up" by progressively adding electrons in orbitals starting at the lowest available, following Pauli exclusion principle and Hund's rule

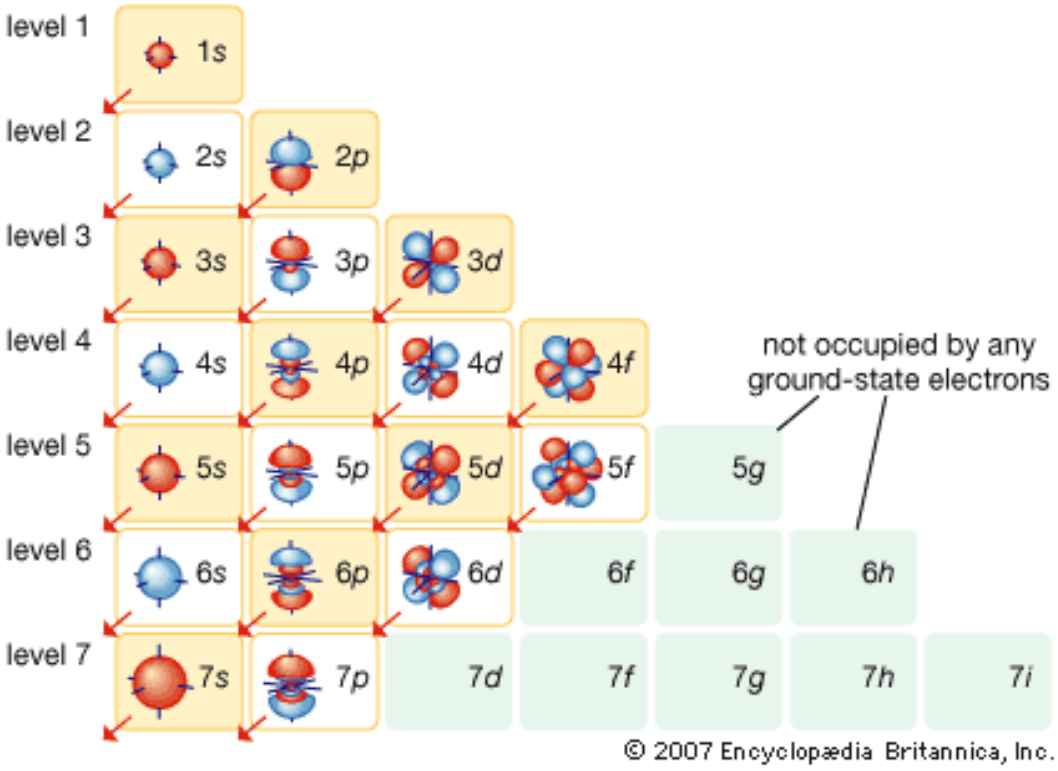


What will be the electronic configuration for Cl? (Z=17)

- **Remember, 2 electrons per s orbital, 6 per p**

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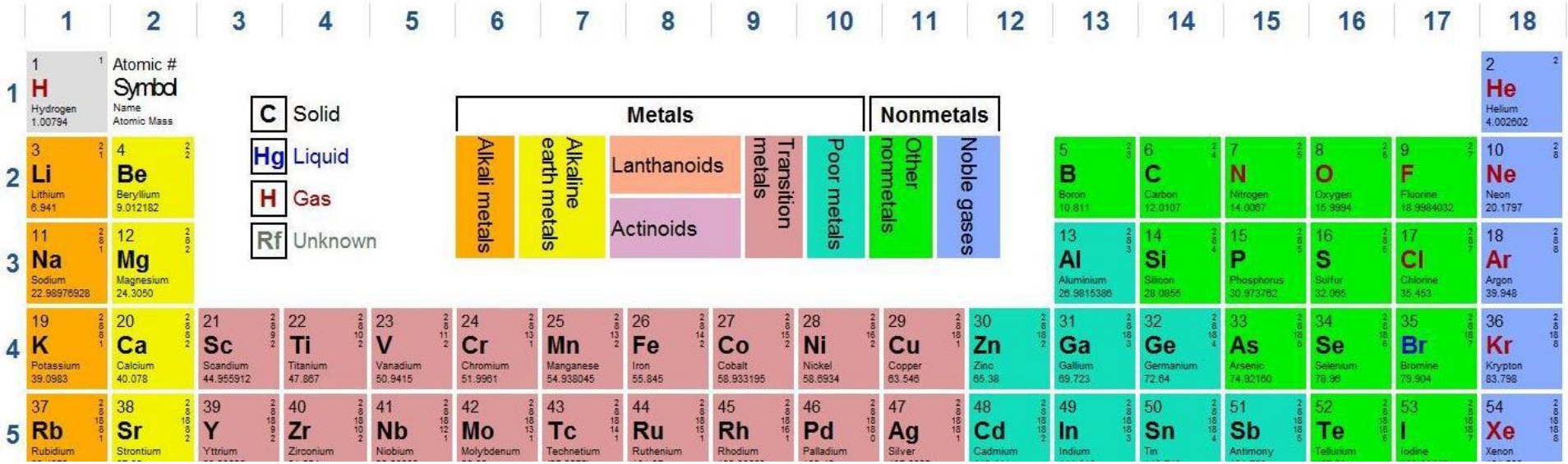


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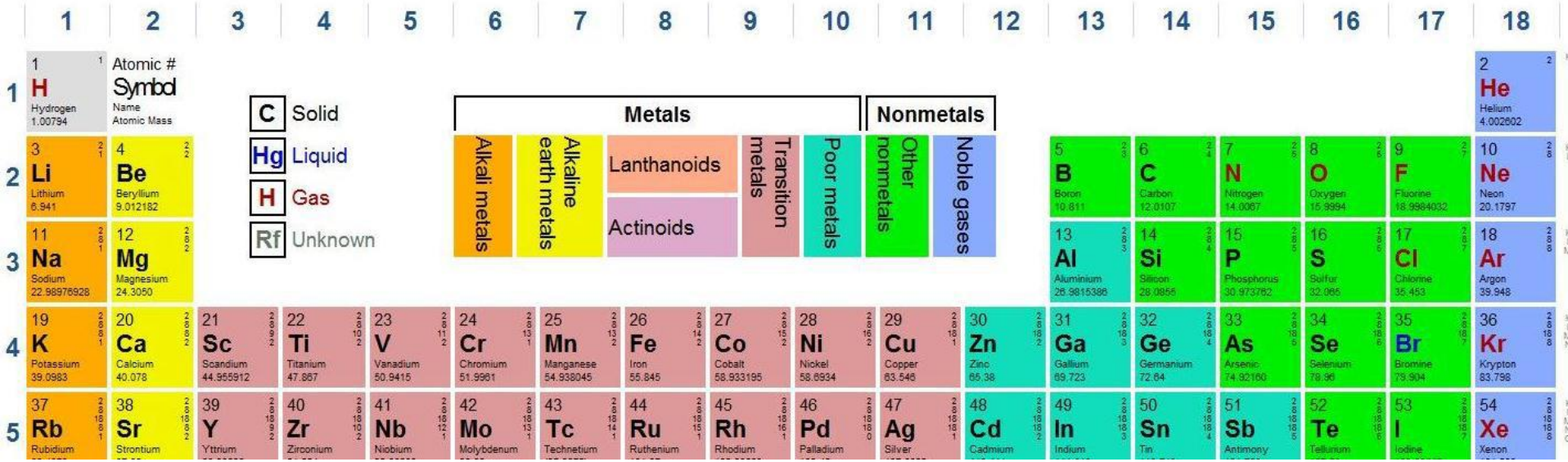
$1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^5$

The Periodic table of elements



1. **Elements in the same row-period have similar size. WHY?**
2. **Elements in the same column-group have similar electronic and chemical properties. WHY?**

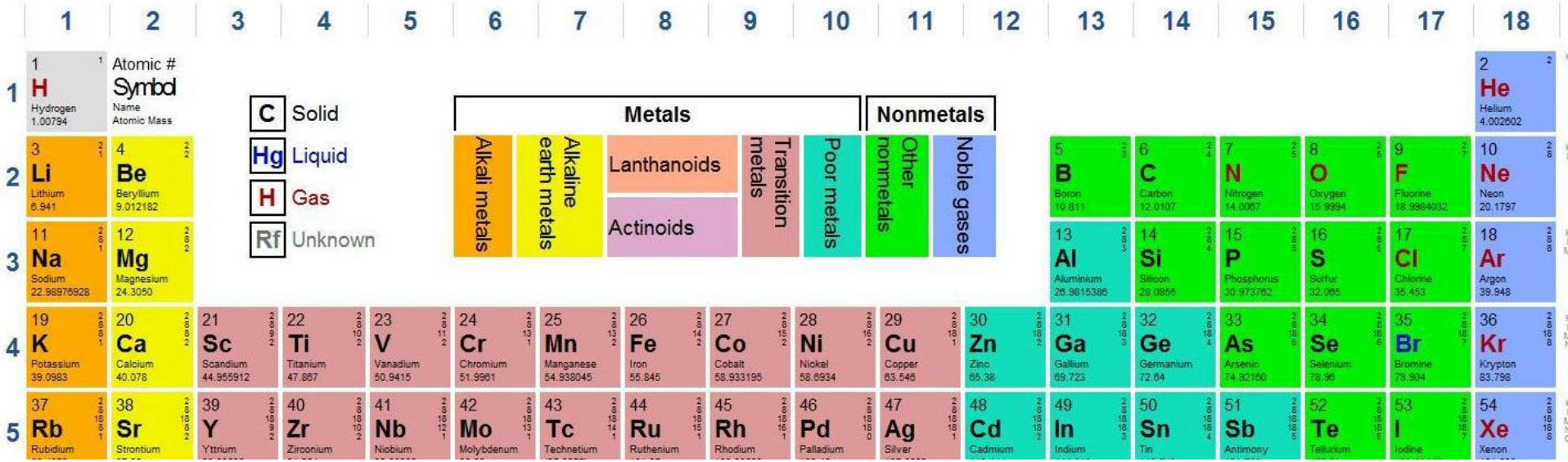
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1. **They have the same number of Atomic Orbitals, *n* number**

The Periodic table of elements

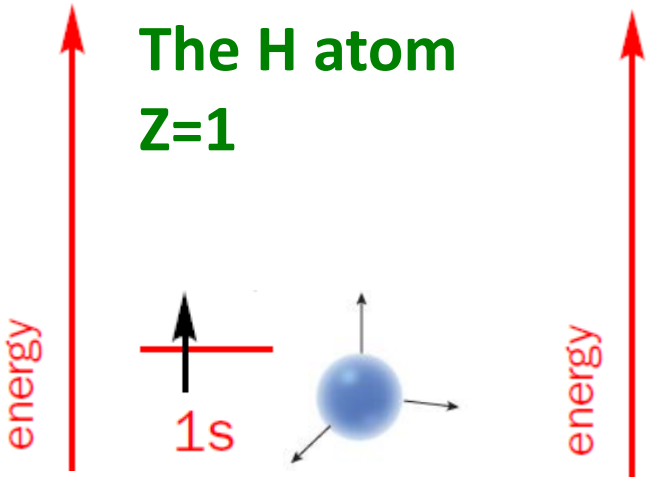


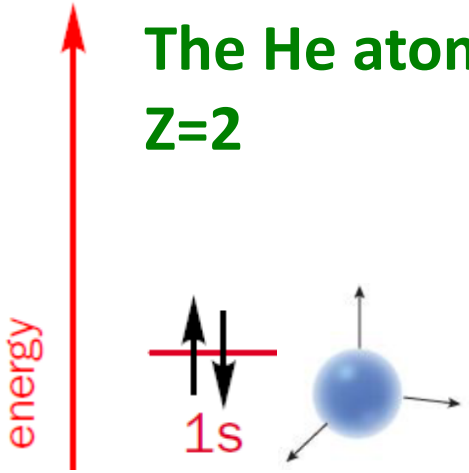
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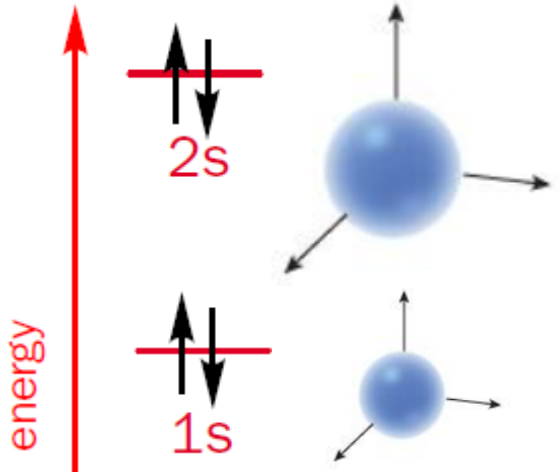
1. **They have the same number of Atomic Orbitals, *n* number**
2. **They have the same number of electrons in their outer Atomic Orbital**

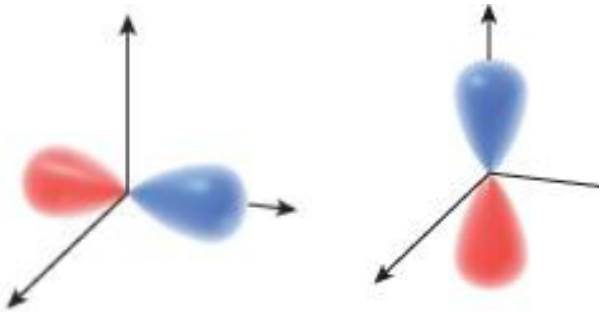
Visit: www.ptable.com

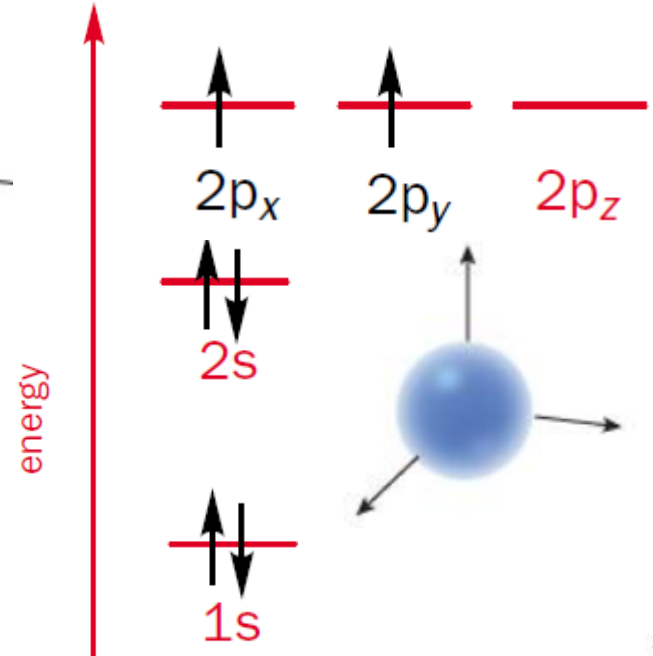
Representing Atomic Orbitals in Energy level diagrams













Structure of Molecules-Summarizing…

- How we know that electrons have different energies?
- How electrons fit into atomic orbitals?
- **How atomic orbitals combine to make molecular orbitals?**
- **Why organic molecules have linear, planar, or tetrahedral structures?**
- **Connection between shape and electronic structure**
- **Picturing the shape and energy of molecular orbitals in simple molecules**
- **Predicting the location of lone pairs and empty orbitals**

What are Molecular Orbitals?

- **When the AOs of 2 atoms combine they form Molecular Orbitals (MOs) that belong to both atoms. Up to 2 electrons can occupy a MO as well. By being negatively charged they exert an attractive force on each of the nuclei and hold them together.**

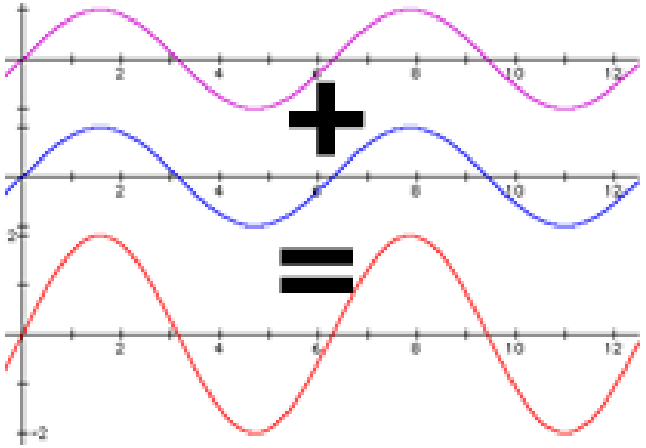
We have a chemical bond!

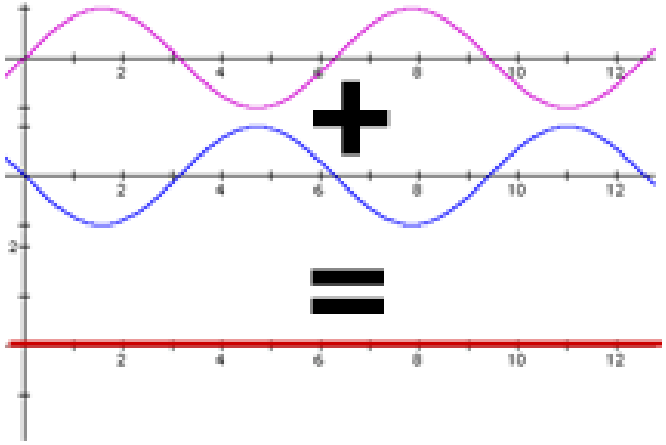
Possible Combinations of Atomic Orbitals

Combination of AOs	MO-Bond formed
s-s	σ (sigma)
p-p	pσ or pπ (pi)
s-p	σ

Combining waves in- and out-of-phase

Electrons: "Wave-particle duality"
Atomic Orbital: Wave function





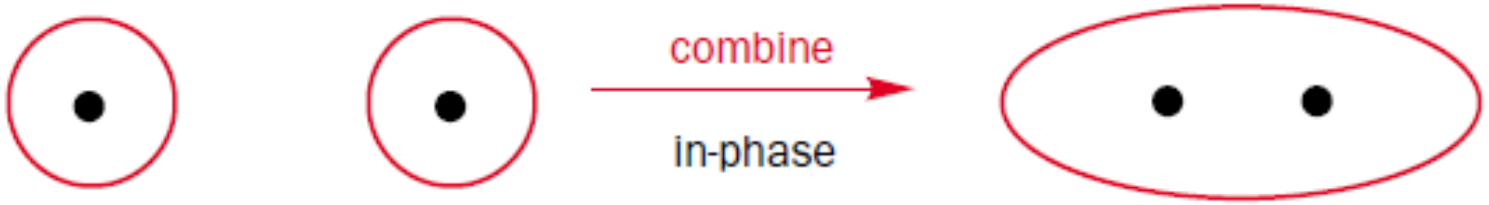
Combining waves in-phase:
Constructive overlap

Combining waves out-of-phase:
Destructive overlap

Bonding Molecular Orbital

Antibonding Molecular Orbital

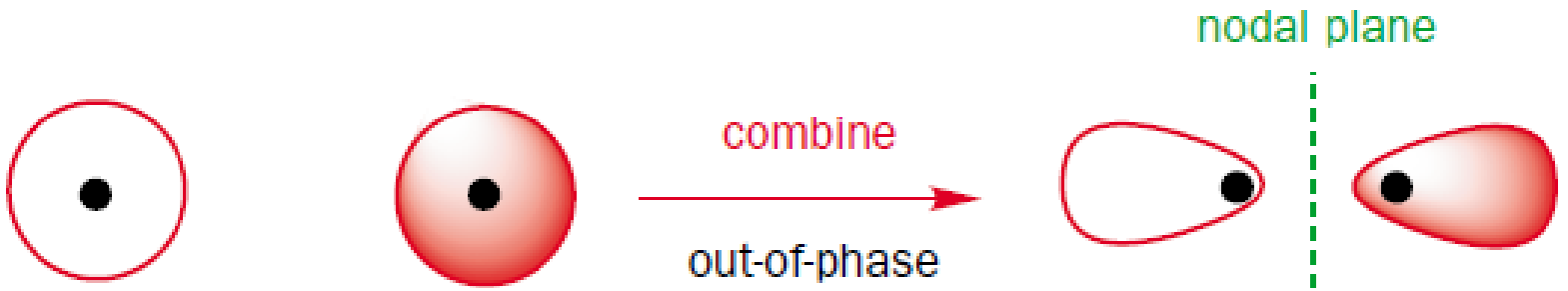
Sigma bonds by interaction of s-s AOs: Not all interactions lead to bond formation!



σ MO

the two 1s orbitals combining in-phase to give a **bonding orbital**

Here we have the formation of a bond

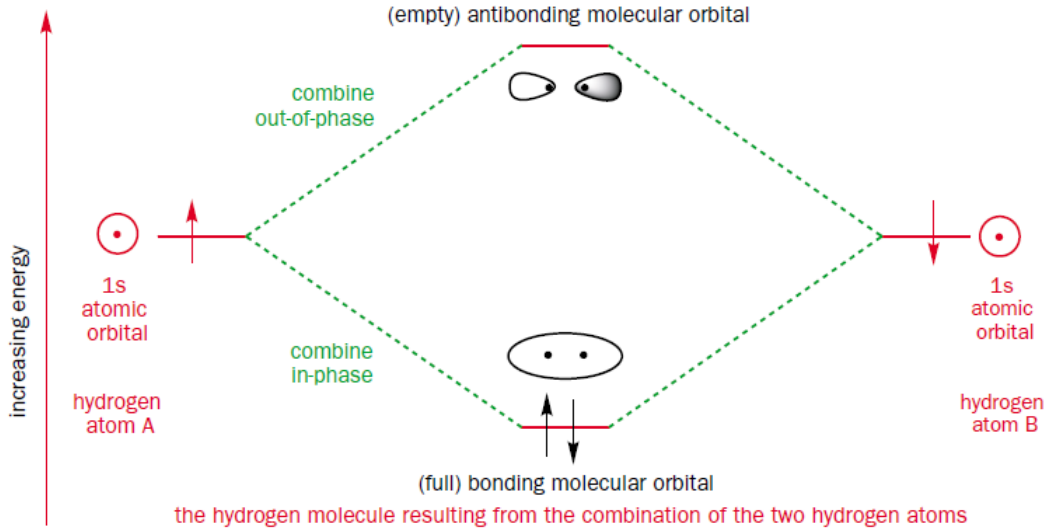


σ* MO

the two 1s orbitals combining out-of-phase to give an **antibonding orbital**

Here we don't!

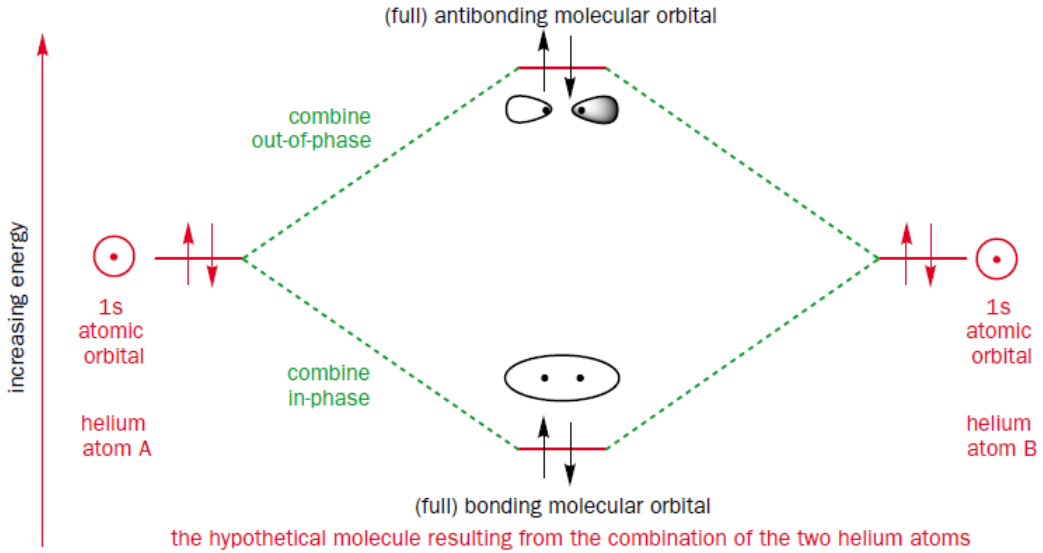
Hydrogen and Helium: Same row, similar size, completely different properties



The molecule of H_2: Bond order: 2-0/2=1

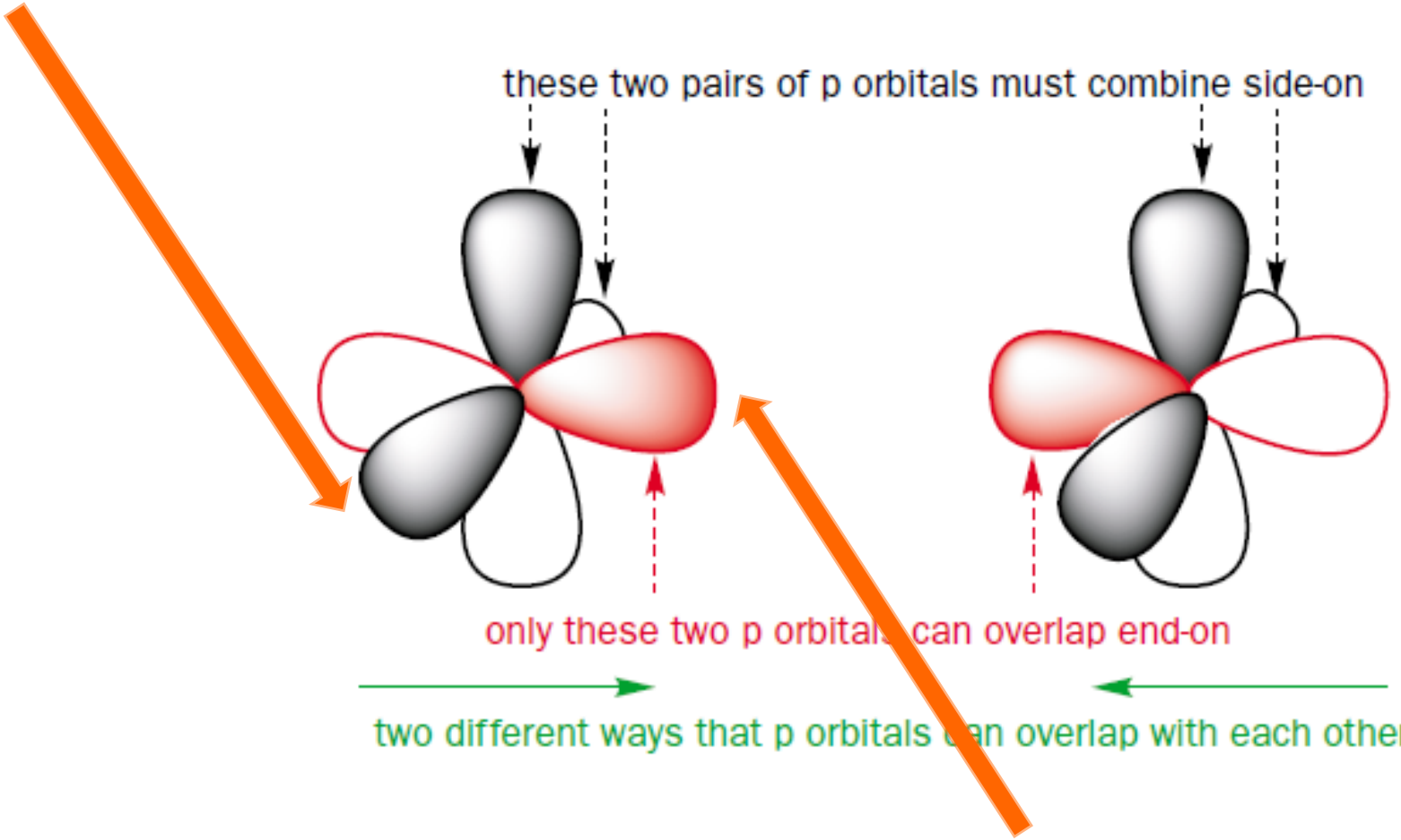
http://www.chemtube3d.com/orbitalshydrogen.htm

The molecule of He_2: Bond order: 2-2/2=0 It does not exist



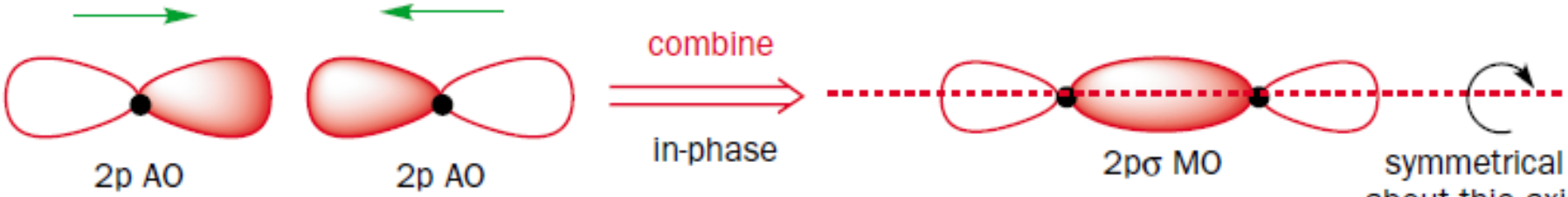
Interaction of p-p AOs: Two possibilities, pσ and pπ bonds

- **Side-on interactions provide pπ bonds**



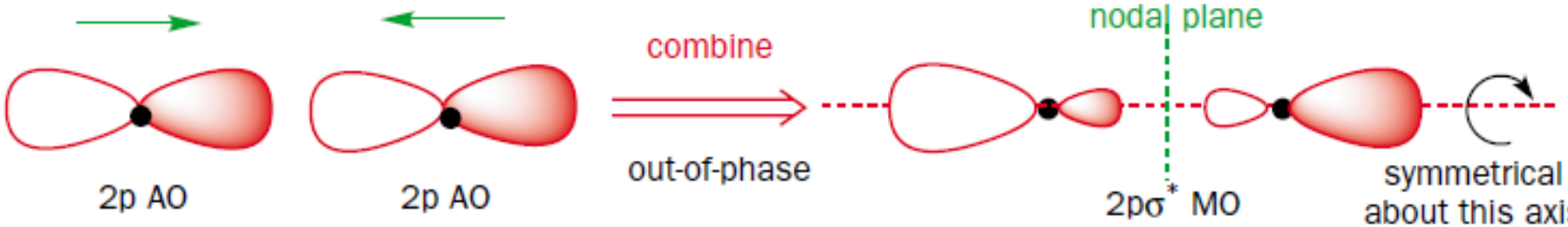
- **End-on interactions provide pσ bonds**

End-on interaction of p-p AOs: The pσ bond



the end-on overlap of two 2p atomic orbitals to give the 2pσ bonding MO

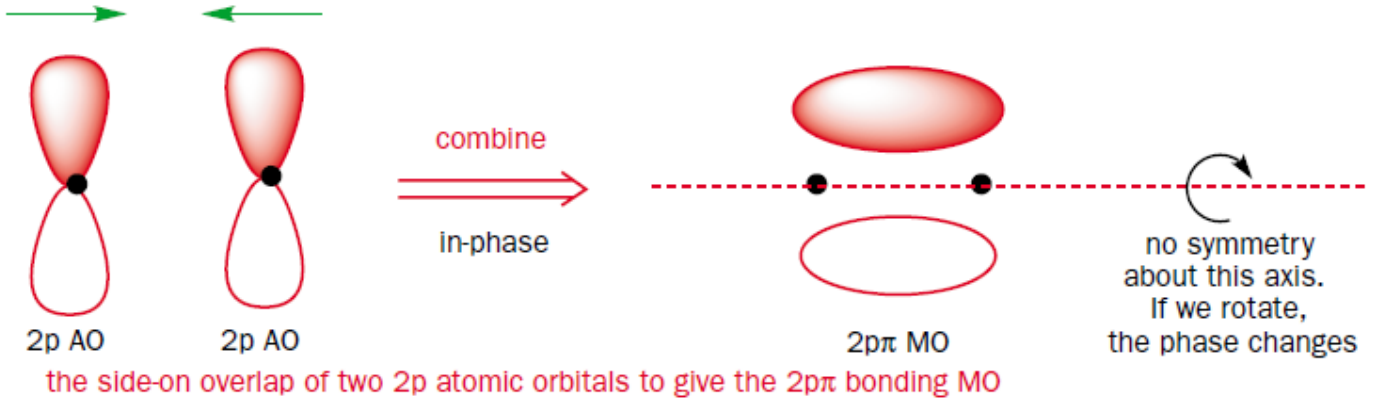
Here we have the formation of a pσ bond



the end-on overlap of two 2p atomic orbitals to give the 2pσ* antibonding MO

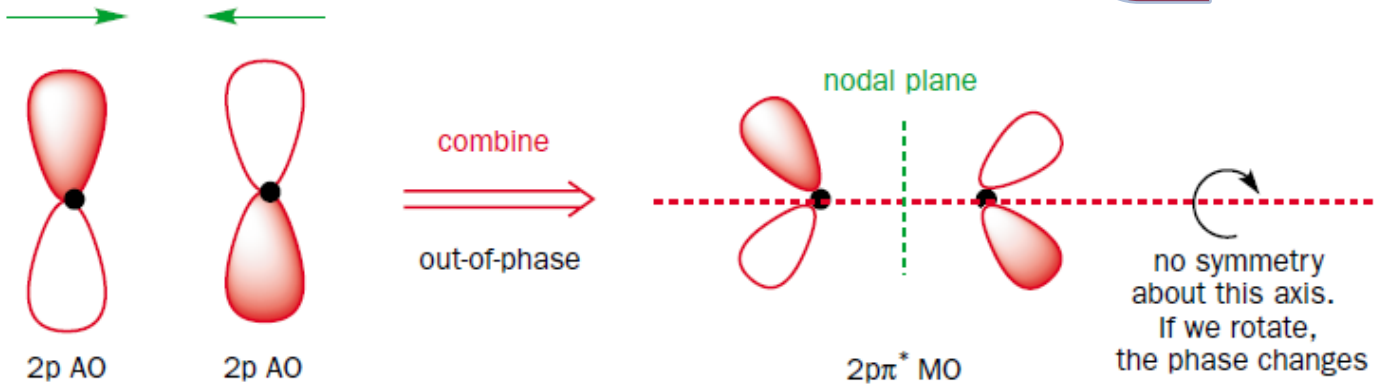
Here we don't!

Side-on interaction of p-p AOs: The pπ bond



the side-on overlap of two 2p atomic orbitals to give the 2pπ bonding MO

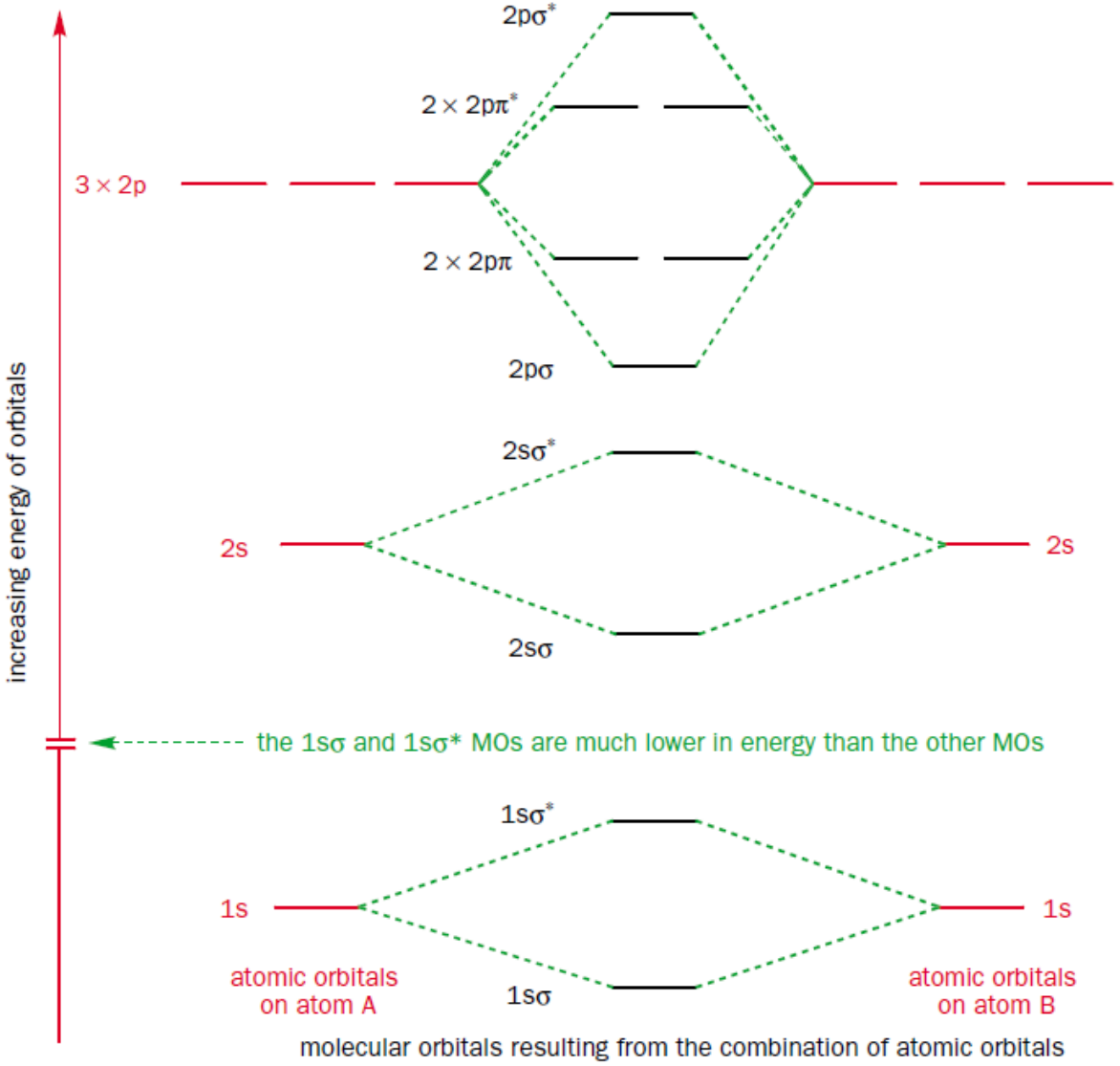
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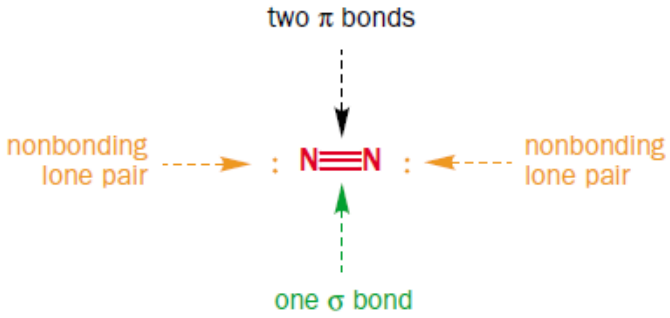


the side-on overlap of two 2p atomic orbitals to give the 2pπ* antibonding MO

Here we don't!

Summary: The s-s and p-p AOs possible combinations



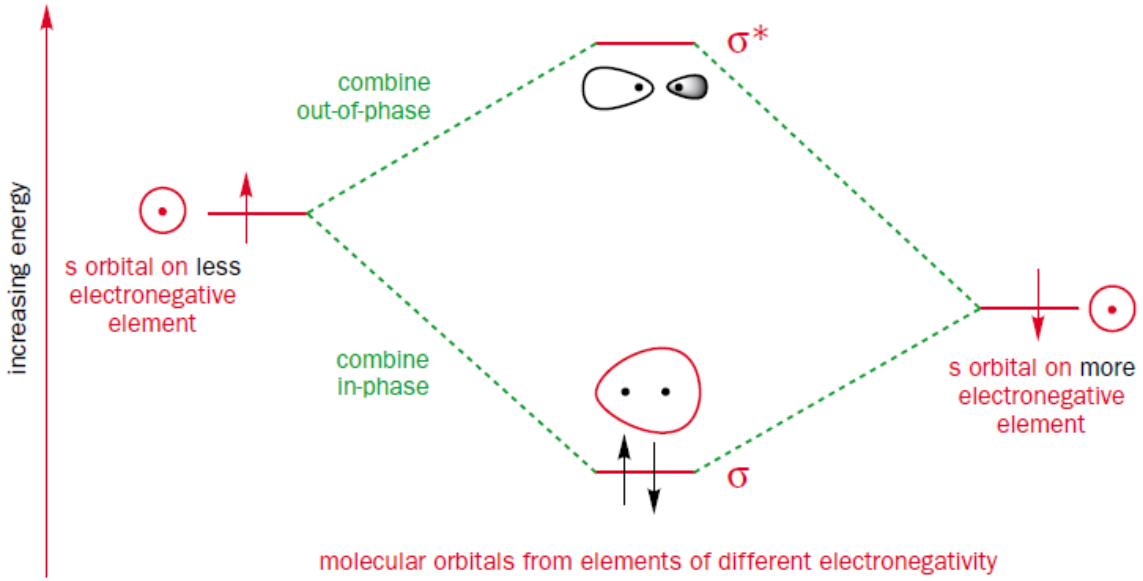


For N: Z=7

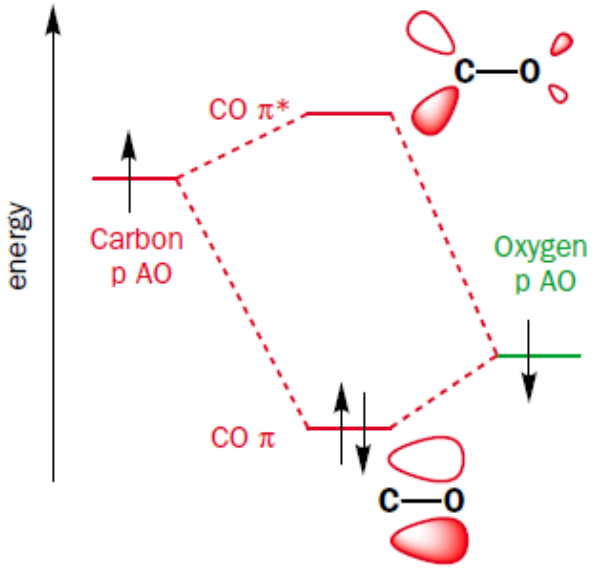
Electronic configuration: 1s2, 2s2, 2p3

Bond order for N_2= (2-2)+(2-2)+(2+2+2)/2=3

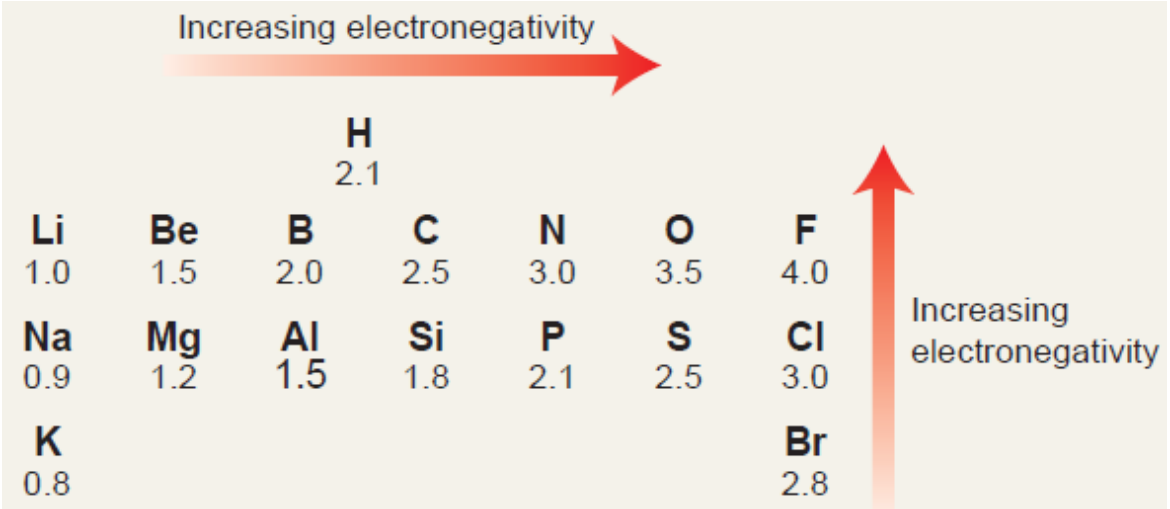
Combining different kinds of AOs: Electronegativity factors



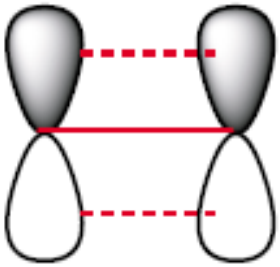
molecular orbitals from elements of different electronegativity



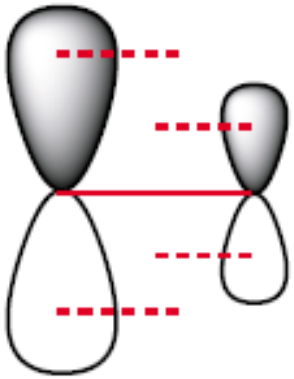
Electronegativity is a measure of an atom's **attraction** for electrons



Combining different kinds of AOs: Size and orientation factors



efficient overlap of p orbitals of the same size (same principal quantum number *n*)



inefficient overlap of p orbitals of different size (different principal quantum numbers *n*)



p_z and p_x

these two p orbitals cannot combine because they are perpendicular to each other



p_z and p_y

here any constructive overlap is cancelled out by equal amounts of destructive overlap

Combining different kinds of AOs: End-on interaction of s-p AOs form a σ bond

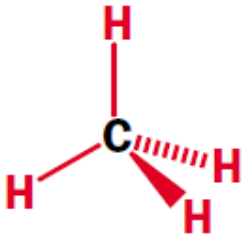


p and s
(side-on)
here any constructive overlap is cancelled out by equal amounts of destructive overlap



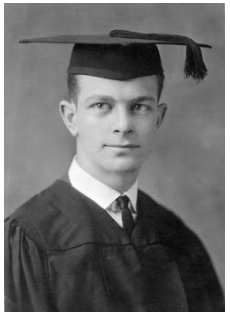
p and s
(end-on)
however, s and p orbitals can overlap end-on

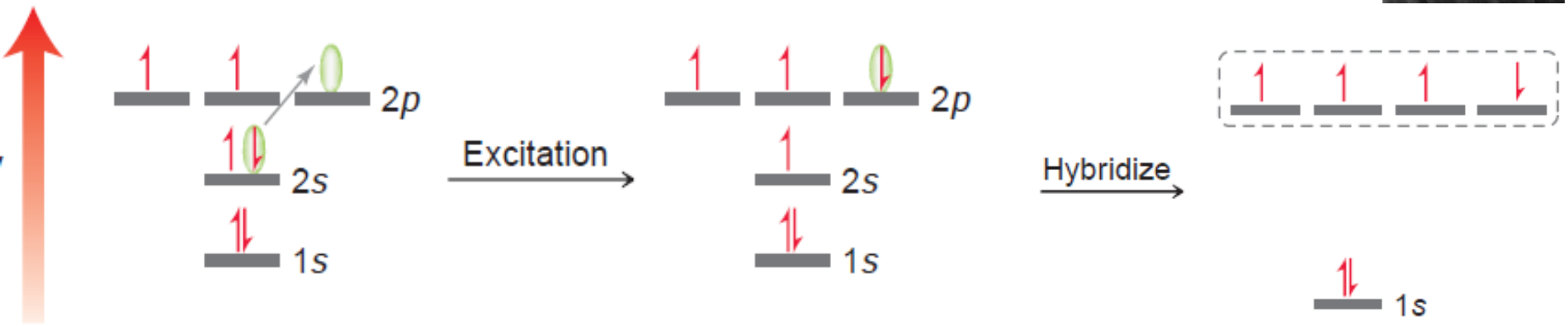
MOs of molecules with more than 2 atoms: Hybridization ⟶ The Methane case



Tetrahedral methane:
C forms 4 equal bonds with 4 H's

Linus Pauling
(1901-1994)
Nobel Prize in Chemistry 1954 for his research into the nature of the chemical bond





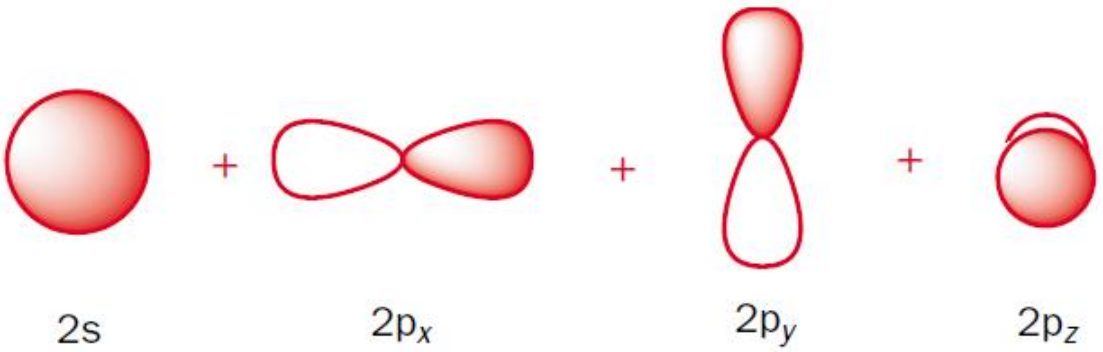
C in ground state:
4 e's, 2s and 2p

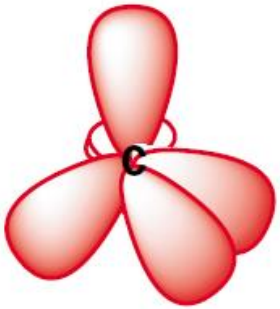
C in excited state:
4 e's, 1s and 3p

Hybridization:
4 e's, 4 sp^3

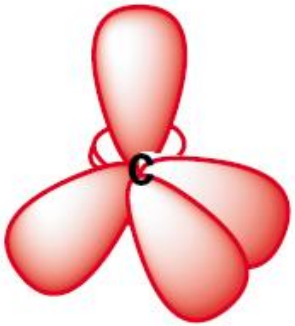
Hybridization is the combination of two or more atomic orbitals to form the same number of hybrid orbitals each having the same shape and energy

sp^3 Hybridization: The Methane case



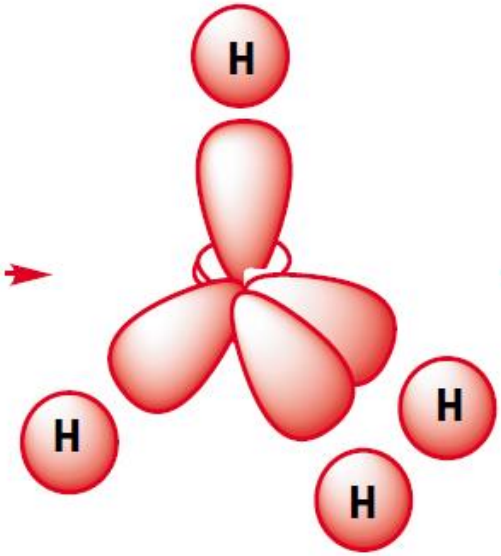


4 sp^3 hybrid orbitals

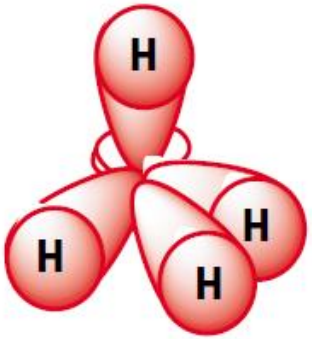


four sp^3 hybrid orbitals form a tetrahedron

add four H atoms



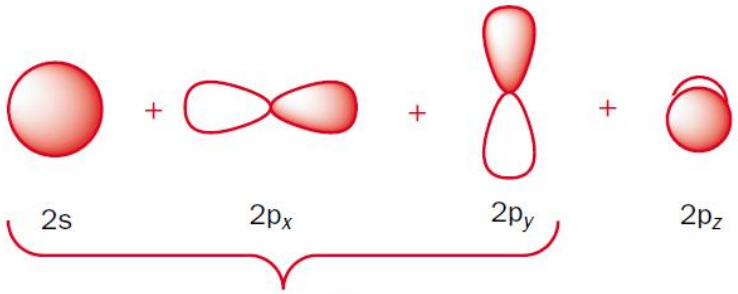
combine sp^3 and 1s

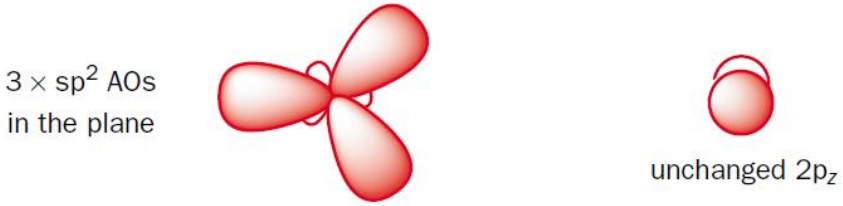


each MO orbital is the same and has σ symmetry

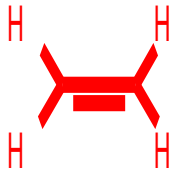
Visit http://www.chemtube3d.com/orbitalshybrid.htm

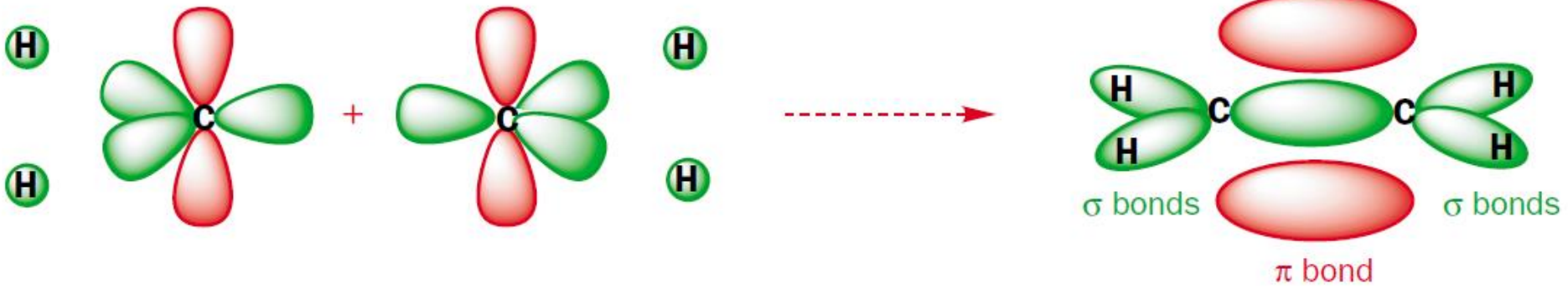
sp^2 Hybridization: The Ethene case





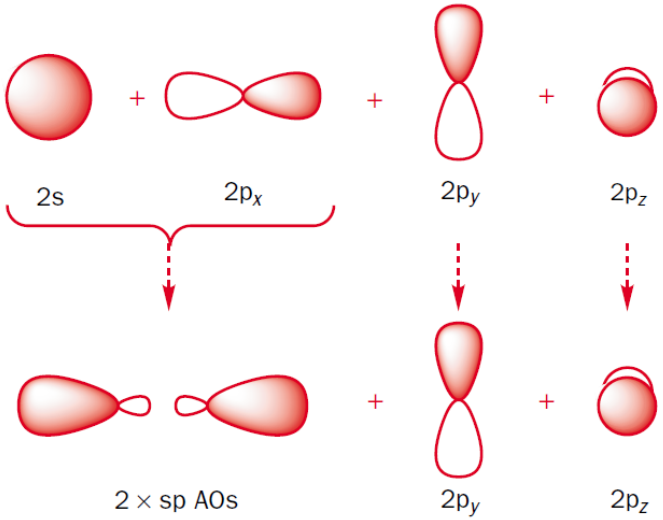
3 sp^2 hybrid orbitals + 1 p unchanged orbital





Visit http://www.chemtube3d.com/orbitalsethene.htm

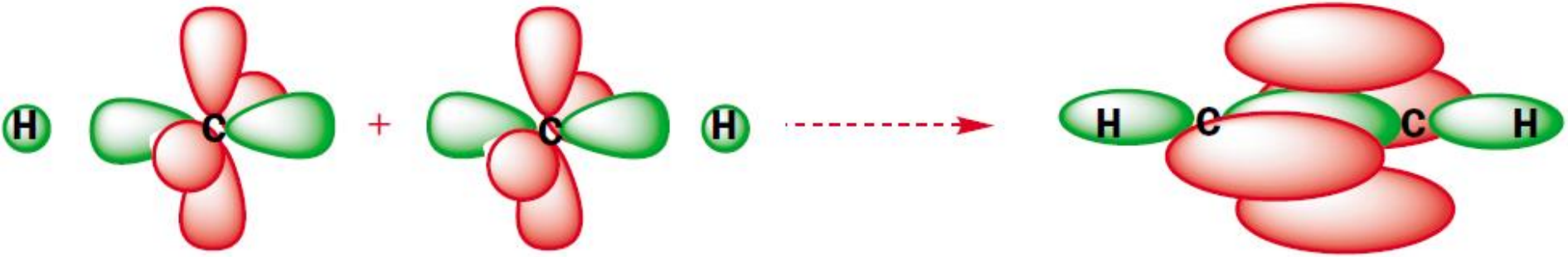
sp Hybridization: The Ethyne case



2 sp hybrid orbitals + 2 p unchanged orbitals



Ethyne

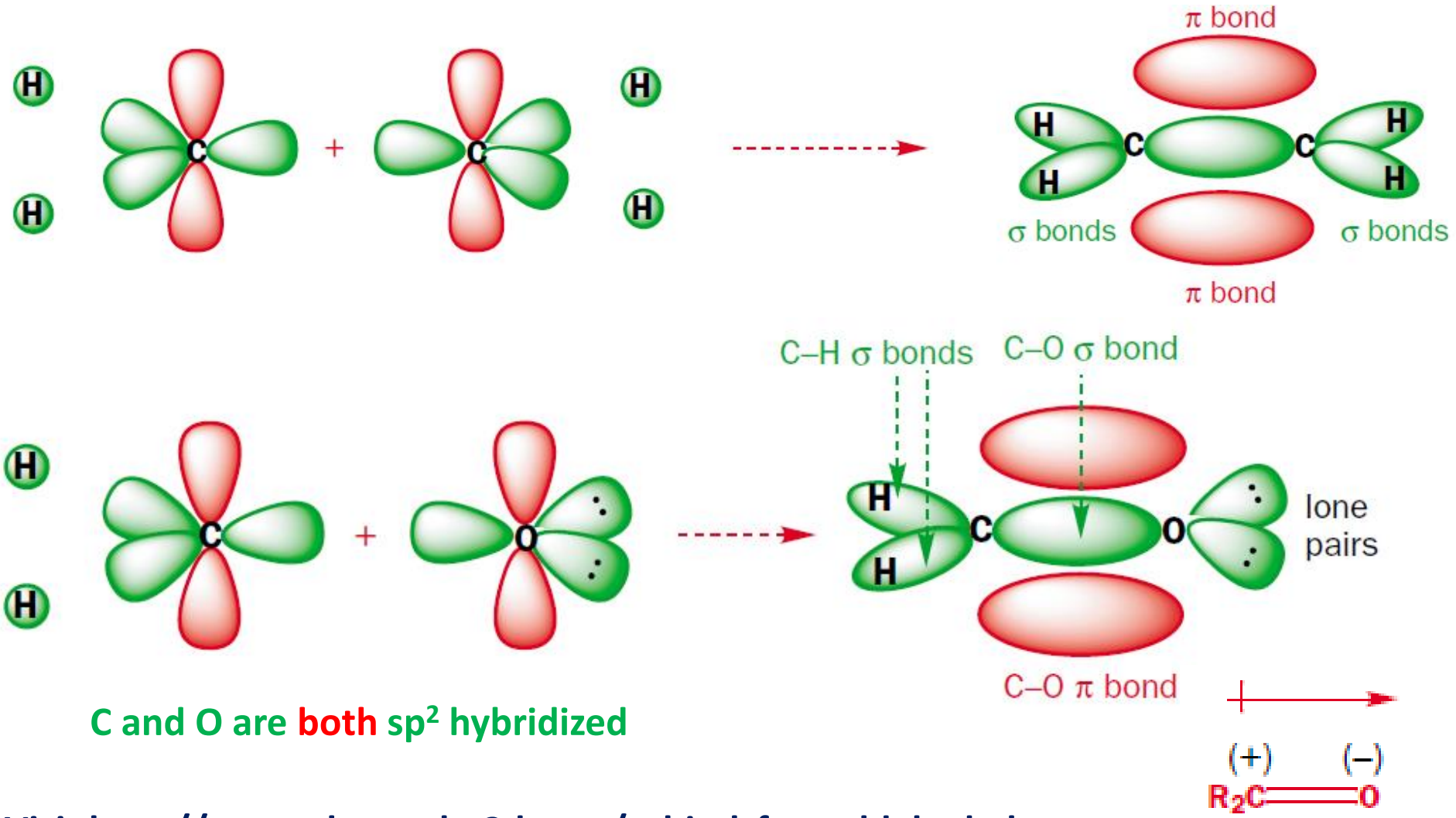


2sp hybridized carbon atoms
2p$_y$ and 2p$_z$ orbitals remain

linear σ bonds form skeleton
two perpendicular π bonds

Visit http://www.chemtube3d.com/orbitalsacetylene.htm

sp^2 Hybridization: Double bonds to other elements



C and O are both sp^2 hybridized

Visit http://www.chemtube3d.com/orbitalsformaldehyde.htm

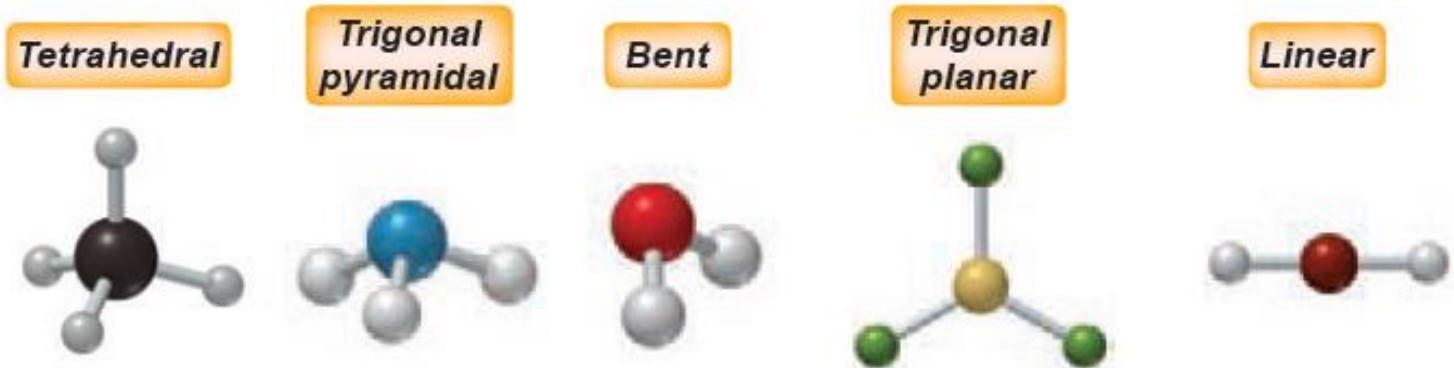
Structure of Molecules-Summarizing…

- How we know that electrons have different energies
- How electrons fit into atomic orbitals
- How atomic orbitals combine to make molecular orbitals
- **Why organic molecules have linear, planar, or tetrahedral structures**
- Connection between shape and electronic structure
- Picturing the shape and energy of molecular orbitals in simple molecules
- **Predicting the location of lone pairs and empty orbitals**

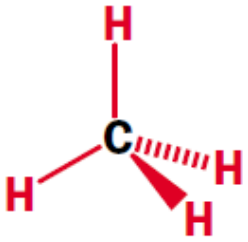
Molecular shape-Bond angle

Nº of groups	Geometry	Bond Angle
4	Tetrahedral	$109{,}8^0$
3	Trigonal planar	120^0
2	Linear	180^0

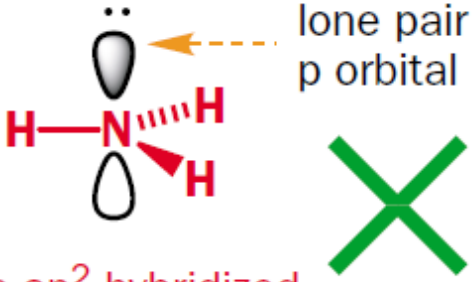
- The number of groups surrounding an atom determines its geometry
- The most stable arrangement keeps these groups as far away from each other as possible
- A group is either an atom or a lone pair of electrons



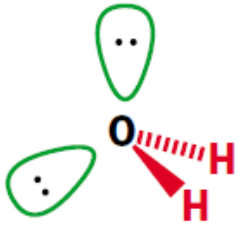
Shapes and electronic structures



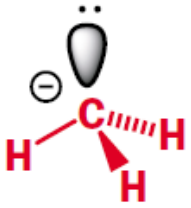
tetrahedral methane
four bonds and no lone pairs



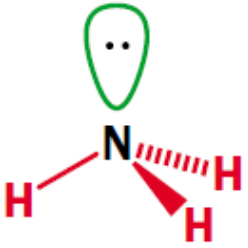
N is sp^2 hybridized



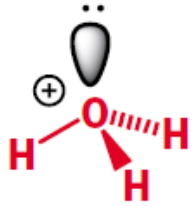
tetrahedral water
two bonds and two lone pairs



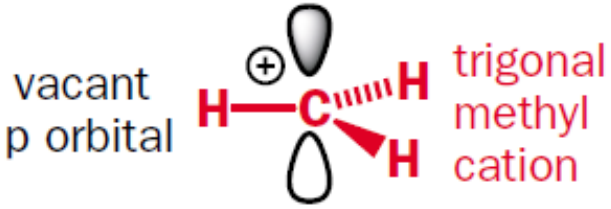
methyl anion

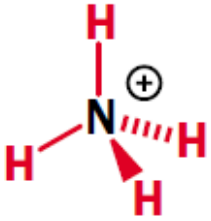


tetrahedral ammonia
three bonds and one lone pair

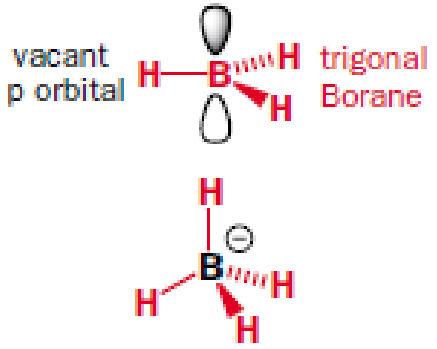


hydronium ion





ammonium cation



borohydride anion

Why and how?

- How does it feel to be a molecule?
- How atoms are held together?
- What is a chemical bond?
- Why molecules have particular shapes?